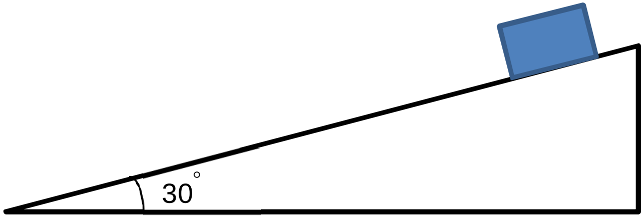
7	A block starts to slide down a 30° slope from rest as shown in the figure below.  Assume that the friction on the block is negligible. What is the final speed of the block after it has travelled 10 m down the slope?						
	A	4.9 m s^{-1}	**B**	8.5 m s^{-1}	**C**	9.9 m s^{-1}	**D**	13 m s^{-1}

Answer: C

Acceleration along the slope = g sin 30° = 4.905 m s^{-1}.

$v^2 = u^2 + 2as$

$v^2 = 0 + 2 \times 4.905 \times 10$

$v = 9.9$ m s^{-1}